

Aaryaman Kompella | Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer (3.76 GPA) with a strong foundation in full-stack development and AI integration. Proven track record of building automated data pipelines, MERN stack web applications, and desktop software utilizing Python, React, and PostgreSQL. Passionate about leveraging data-driven architectures to optimize workflows and drive technical efficiency.

EDUCATION

Bachelor's in Science in Computer Science (Expected May 2027)

Stony Brook University

TECHNICAL SKILLS

- **AI & Data Science** | Python, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, Machine Learning, Exploratory Data Analysis (EDA).
- **Software & Platform** | Java, C, OCaml, JavaScript, React, PostgreSQL, MySQL, Git/DevOps.
- **Productivity & Design** | MS Office, Figma, Fusion 360.
- **Languages** | Native fluency in English, Telugu, Hindi, and Gujarati; Intermediate Japanese.

EXPERIENCE

AI Workforce & Development Intern | SUNY Research Foundation

July 2026 – September 2026

- Engineered a **centralized data pipeline** to standardize New York State Department of Labor (NYSDOL) apprenticeship crosswalks across **30+ SUNY campuses**, eliminating redundancies in hundreds of templates and **streamlining the active course catalog by 40%**.
- Developed a **standalone desktop application** using **Python (Tkinter, Pandas)** equipped with **dynamic multi-criteria filtering**, full-database search, and algorithmic color-coded match scoring to enable seamless **offline data validation** for campus administrators.
- Architected a cloud-based AI pipeline concept utilizing **Azure OpenAI** and a **PostgreSQL vector database** to automatically extract and map unstructured SUNY coursework to strict DOL training requirements, projecting a **70% reduction in manual verification time**.
- Analyzed consolidated crosswalk data to identify high-demand apprenticeship trades, guiding **strategic funding allocation** for corporate partnerships and expanding career pipelines for **over 500 students annually** (5,000+ total program completions).

Web Development Intern | SSC Infotech LLP

June 2024 - August 2024

- Developed a clean, user-friendly interface for a client website supporting **200+ daily users**, focusing on technical requirements and user experience.
- Utilized CSS to engineer a fully responsive interface across all device types, implementing custom slick sliders to enhance modern UI aesthetics.

Resident Assistant | Stony Brook University

August 2024 - Present

- Facilitate **conflict resolution** among 35+ residents, while **mentoring** 7 council members as a Hall Council Liaison, providing leadership and program guidance.
- Coordinate with a 7-person team to oversee 250+ residents, ensuring a supportive and organized environment through effective teamwork.

PROJECTS

Data Mapping & Classification (Python, Scikit-Learn, Pandas)

October 2025

- Engineered an **AI-driven** decision tree classifier, optimizing depth and split parameters using **GridSearchCV** to analyze a credit card transactions dataset of over **1 million entries**.
- Conducted **Exploratory Data Analysis (EDA)** using **Seaborn** to construct heatmaps and scatterplots, isolating highly correlated variables like the ratio to median purchase price to inform the machine learning architecture.
- Tuned the classification model utilizing **cross-validation** and **balanced class weights** to address a significant class imbalance (8.7% fraud), achieving an F1-score of 0.84 and a **79% fraud detection rate** without overfitting.

Playlist Web App (JavaScript, MongoDB, SQL, Postgres)

September 2025

- Developed a full-stack MERN (MongoDB, Express, React, Node.js) web application.
- Engineered a back-end RESTful API with Express to manage data in a MongoDB database.
- Built a dynamic, component-based React front-end featuring client-side routing and secure user authentication.
- Implemented CRUD operations, including undo/redo functionality, for user-generated playlists and songs.
- Engineered a supplementary Python and PostgreSQL microservice to securely manage user records alongside the primary MERN application.

EXTRACURRICULAR

Canon USA Mentorship | Stony Brook University

August 2024 - June 2025

- Collaborate with a mentor from Canon USA to learn corporate strategies and project development cycles.
- Gained exposure to projects involving **Machine Learning and Cloud technologies**, understanding the integration of various departments in technical workflows.